

Date: Jan 25, 2021

Experiment: 2

Title: Determination of melting point (mp)

Objective

The objective of this experiment was to determine the melting point (mp) of the sample we recrystallized in the first experiment (the benzoic acid)

Theory

The determination of a melting point of a sample is conducted using a capillary tube. . The first is a manual apparatus setup with the aim of heating up a sample inside a capillary tube whilst visually looking for when the sample has melted. The concept involves placing a small amount of the substance in a capillary tube and joining this to a thermometer. The capillary tube is then placed into a heating bath and the experiment is conducted. It is used to identify a sample, and establish its purity. A pure sample has a sharp melting point with a very small range while an impure substance has a lower melting point with a broad range. A melting range is the difference between the temperature at which the sample begins to melt and the temperature at which the sample has actually melted.

Procedure

A small amount of the dry benzoic acid we recrystallized last week was put into a capillary tube by checking the sealed down end. This capillary tube was dropped into a long glass tube, and as a result a small amount of benzoic acid was inside the capillary tube. Then the capillary tube and a thermometer were held together by a small rubber band. The thermometer which was held together with the capillary tube was immersed into an oil bath in a test tube (a test tube was used instead of a Thiele tube) by making sure the rubber band was above the oil. The test tube was held by a clamp and at the bottom it was heated by a Bunsen burner. The oil started boiling, and the sample inside the capillary tube was starting to melt with the indirect heat originating from the boiling oil. When the thermometer read 100°C the test tube was heated in a slower rate, and this process was continued to avoid missing the temperature it started to melt. The temperature started rising, and at 120°C the sample inside the capillary tube started melting. The temperature it started melting, and the temperature it finished melting were recorded carefully by observing the thermometer at the same time watching the sample. The melting point was recorded as a range.

Apparatus

- Capillary tube

- Glass tube
- Thermometer
- Rubber band
- Bunsen burner
- Clamp
- Oil bath

Chemicals used

- Benzoic acid

Result and discussion

Compound	Melting point
Benzoic acid (C ₉ H ₆ O ₂)	Starting temp 120°C-ending temp 122°C

The melting point of a substance will determine the purity of a substance. For benzoic acid, the result of the melting point was 120°C-122°C and it was within the range of the accepted values. Even if the observed value is closer to the accepted value, the pure compounds were seen to have a narrow range, while impurities were seen to cause the melting range to become lower and broader.

Quiz questions

1. Why is mp often reported as range and not as exact figure?

- Pure crystalline compounds usually have a sharp melting point. That is, the melting point range-the difference between the temperature at which the sample begins to melt and the temperature at which the sample is completely melted-is small (narrow). This is because each substance acts as an impurity in the other. Impurities, even when present in small amounts, usually lower the melting point and broaden the melting point range. A narrow melting point range (0.5-2°C) usually indicates that the substance is fairly pure. To summarize, if a crystalline substance is pure, its melting point range is likely to be narrow. So due to an impurity often melting point is recorded as a range.

2. Why does an impurity reduce the mp of a substance?

- Impurities in a crystalline solid disrupt the repeating pattern of forces that holds the solid together. Therefore, a smaller amount of energy is required to melt the part of the solid surrounding the impurity. An impurity as stated above disrupts the crystal lattice forces and less thermal energy is needed to make the compound melt. The impurity also increases the melting point range since the compound is not homogeneous and different areas will start to melt before purer areas in the solid.



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COLLEGE OF HEALTH SCIENCE

DEPARTMENT: PHARMACY

ORGANIC CHEMISTRY LAB

REPORT: EXP 2

GROUP MEMBERS:

- 1. MERON ASFAW: UGR/0336/12**
- 2. MEZID MUHAMMED: UGR/9636**
- 3. MELAT SALLE:UGR/1574/12**

SUBMITTED TO: INSTUCTOR KIBROM GEBREHIWOT

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